

# MAD-1 Project Report

## Author

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**About Me :** I am pursuing two undergraduate degrees simultaneously: a BS in Data Science and Applications at IIT Madras (currently in the Diploma Level) and a B.Tech in Computer Science and Engineering at BIT Sindri, Dhanbad (currently in the 4th semester).

## Project Details

**Project Title :** Trekking Management System

### Problem Statement :

To design and develop a role-based web application that enables administrators, trek staff, and trekkers to efficiently manage trekking activities, bookings, staff assignments, and trek operations.

### Approach :

The application was developed using Flask with a modular Blueprint-based architecture. SQLAlchemy was used for database operations, while Jinja2 templates rendered the frontend. The system enables administrators to manage treks, staff, trekkers, and bookings; staff members to manage assigned treks; and trekkers to browse and book available treks

## AI/LLM Declaration

I used ChatGPT (GPT-5) and Claude Sonnet 5 to assist in discussing the database schema, application workflow, routing, HTML/Jinja template design, debugging, and understanding Flask concepts.

Approximately **50–60%** of the development involved AI-assisted discussions and code suggestions. All final implementation decisions, code integration, testing, and debugging were carried out manually after understanding the generated suggestions.

## Technologies and Frameworks Used

| Technology/Library | Purpose                                                |
|--------------------|--------------------------------------------------------|
| Flask              | Core backend web framework                             |
| SQLite             | Lightweight relational database for persistent storage |
| Flask-SQLAlchemy   | ORM for database modeling and queries                  |

|                            |                                                  |
|----------------------------|--------------------------------------------------|
| <b>HTML</b>                | Structuring web pages                            |
| <b>Jinja 2</b>             | Template engine for rendering dynamic HTML pages |
| <b>CSS and Bootstrap 5</b> | Frontend styling and responsive design           |
| <b>Matplotlib</b>          | Generating analytical charts for admin dashboard |

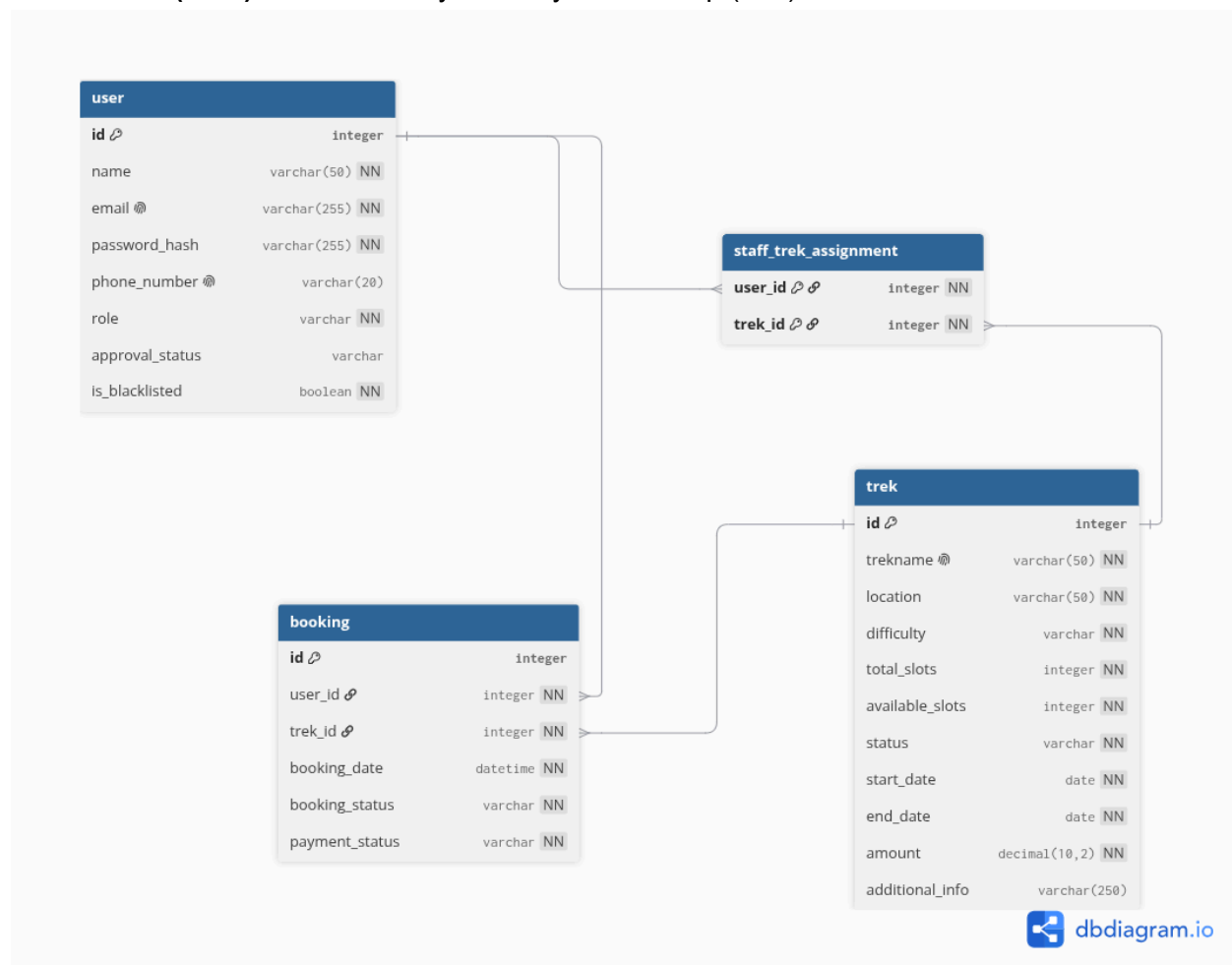
## Database Schema / ER Diagram

### Tables :

1. **User** : stores user details (id, name, email, password\_hash, phone\_number, role, approval\_status, is\_blacklisted)
2. **Trek** : stores trek details (id, trekname, location, difficulty, total\_slots, available\_slots, status, start\_date, end\_date, amount, additional\_info)
3. **Booking** : stores booking details (id, user\_id, trek\_id, booking\_date, booking\_status, payment\_status)
4. **StaffTrekAssignment** : stores staff assignment to treks (user\_id, trek\_id).

### Relationships :

1. **User (trekker) to Booking** : one-to-many relationship (1:m)
2. **Trek to Booking** : one-to-many relationship (1:m)
3. **User (staff) to Trek** : many-to-many relationship (m:n)



## API Design

No REST/JSON API was implemented for this project — all core requirements were built using traditional Flask routes and server-rendered Jinja2 templates.

## Architecture & Features

### Architecture Overview :

- **app.py** – Flask application entry point
- **/application** – models, database configuration, utilities and decorators
- **/application/controllers** – Blueprint modules implementing application routes
- **/templates** – Jinja2 templates
- **/static** – CSS stylesheets and generated chart images

### Implemented Features :

- **User Authentication:** Login for the admin; registration and login for staff and trekkers.
- **Admin Approval System:** Admin can approve or reject staff registration requests, and blacklist/un-blacklist staff and trekkers.
- **Admin Trek Management:** Create and edit treks, update trek status, and manage staff assignments.
- **Admin Booking Management:** Approve or reject booking requests, and cancel approved bookings with automatic refunds.
- **Staff Trek Management:** Update trek capacity and change the status of assigned treks (start, complete, or cancel).
- **Trekker Management:** Browse available treks, submit booking requests through a dummy payment page, cancel eligible bookings, and view booking history.
- **Admin Dashboard & Analytics:** View system statistics and visual charts, including popular treks, booking trends, and staff/trekker distributions.
- **Search:** Search treks, staff and trekkers by ID or name.
- **Profile Management:** Staff and trekkers can view and update their profile details.

## Video Presentation

### Drive Link :

[https://drive.google.com/file/d/1XzXlvd4Pccz\\_8wAadcJS-vLRkqRs11Rp/view?usp=drive\\_link](https://drive.google.com/file/d/1XzXlvd4Pccz_8wAadcJS-vLRkqRs11Rp/view?usp=drive_link)